

AMAN KUMAR PANDA

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PROFESSIONAL SUMMARY

Cybersecurity analyst and Master of Computer Applications candidate specializing in offensive security, digital forensics, threat intelligence, and network detection and response. Hands-on experience in vulnerability assessment, penetration testing, network security, and machine learning-based anomaly detection. Developer of PhantomEye, an AI-powered OSINT framework, and NeuroShield NDR. Seeking to contribute technical expertise to security operations, threat intelligence, vulnerability research, or NDR engineering roles.

CORE COMPETENCIES

Network Security | Threat Intelligence | Incident Response | Vulnerability Assessment | Penetration Testing | OSINT | Web Application Security | Network Detection and Response | Digital Forensics | Malware Analysis | Risk Assessment

TECHNICAL SKILLS

Security Tools:	Burp Suite, Nessus, Nmap, Wireshark, Metasploit Framework, OWASP ZAP, OpenVAS, FTK Imager, Maltego, Kali Linux, Parrot OS
Networking:	TCP/IP, DNS, HTTP/HTTPS, Packet Analysis, Flow Reassembly, Network Protocols, Subnetting, VLANs
Frameworks & Standards:	OWASP Testing Guide, OWASP Top 10, NIST Cybersecurity Framework, ISO 27001, CVSS Scoring, SDLC

PROFESSIONAL EXPERIENCE

Cybersecurity Intern & Professional Trainee

07/2024 – 12/2024

Hacker School (A Unit of Cartel Software)

- Completed 200+ hours of intensive training in ethical hacking, threat analysis, vulnerability assessment, and incident response methodologies.
- Performed manual and automated penetration tests on web applications using Burp Suite and OWASP ZAP; identified and documented SQL injection, XSS, CSRF, and authentication bypass vulnerabilities with CVSS scoring.
- Conducted network reconnaissance and vulnerability scanning using Nmap and Nessus across lab environments; analyzed scan results to prioritize critical findings and recommend remediation strategies.
- Utilized Metasploit Framework to simulate real-world attack vectors, validate defensive controls, and support remediation planning.
- Analyzed network traffic using Wireshark to detect anomalies and potential indicators of compromise in simulated enterprise environments.

Operations & HR Intern

06/2024 – 09/2024

IKKASA

- Managed end-to-end recruitment operations and onboarding for 15+ new hires, ensuring accurate employee records and streamlined documentation workflows.
- Coordinated cross-functional administrative activities and supported founders with operational process optimization.
- Developed standardized documentation templates that improved onboarding efficiency and reduced administrative overhead.

PROJECTS

NeuroShield NDR — AI-Driven Network Detection & Response System

Python, scikit-learn, scapy, FastAPI, NumPy, pandas, WebSocket | github.com/349100/NeuroShield-NDR

- Architected and developed an AI-powered Network Detection and Response (NDR) system that captures network traffic, extracts 13 statistical features, and detects anomalies using hybrid machine learning (Isolation Forest + Random Forest).
- Implemented real-time packet capture and flow reassembly using scapy with 5-tuple tracking, timeout-based expiration, and thread-safe state management for live network monitoring.

- Engineered feature extraction pipeline computing Shannon entropy, inter-arrival times, packet size distributions, TCP window sizes, and DNS query frequencies for behavioral threat detection.
- Built automated response engine with severity-based decision matrix (Critical, High, Medium, Low) mapping to quarantine, ACL block, SOC alert, and log-only actions.
- Developed FastAPI REST backend with WebSocket support for real-time dashboard updates, serving JSON threat data and live metrics to an HTML5 analytics frontend.
- Created baseline retraining script that captures real network traffic for 24+ hours and retrains the anomaly detector on user-specific normal behavior to minimize false positives.

PhantomEye — AI-Powered OSINT Intelligence Framework

Python, AsyncIO, REST APIs, DNS, Web Scraping | github.com/349100/phantomeye

- Architected a modular, asynchronous OSINT framework with plugin-based architecture supporting domain reconnaissance, social media enumeration, web intelligence, and threat intelligence modules.
- Integrated OpenAI LLM API for automated correlation analysis, pattern recognition, and natural language generation of executive security summaries.
- Built robust async HTTP client with rate limiting, retry logic, and proxy support to ensure ethical, compliant passive reconnaissance operations.
- Developed automated report generation engine outputting findings in JSON, CSV, and HTML formats with severity-based visual categorization.

SecProbe — AI-Assisted Web Security Misconfiguration Scanner

Python, Flask, REST API, Google Gemini, Groq LLaMA | github.com/349100/secprobe

- Developed an automated web security scanner that detects missing HTTP security headers, SSL misconfigurations, and exposed ports with CVSS-style risk scoring.
- Integrated free AI providers (Google Gemini, Groq LLaMA) for intelligent vulnerability explanation, impact assessment, and remediation guidance generation.
- Built Flask REST API and CLI interface supporting JSON and HTML report export with dark-themed dashboard for security assessment visualization.

Parking Management System — Security-Focused Full-Stack Application

PHP, MySQL, HTML/CSS, JavaScript, Apache

- Designed and developed a full-stack parking management system implementing secure user authentication, session management, and role-based access control.
- Mitigated SQL injection and cross-site scripting vulnerabilities through parameterized queries, input sanitization, and output encoding. Applied secure software development lifecycle principles from requirements gathering and threat modeling to secure deployment.

LEADERSHIP & ACTIVITIES

- **Head Boy:** Elected school Head Boy demonstrating leadership, organizational responsibility, and cross-functional team coordination.
- **Lead Graphic Designer:** Led a creative design team to produce promotional materials for university events, managing timelines and deliverables.

EDUCATION

Master of Computer Applications (MCA) — Cybersecurity & Forensic Science

Parul University, Vadodara, Gujarat | 08/2025 – 07/2027 | Expected Graduation: July 2027

Bachelor of Computer Applications (BCA)

Sambalpur University, Odisha | 08/2022 – 07/2025

CERTIFICATIONS

- Certified Ethical Hacker (CEH) v12 (Completion)
- Certified Cybersecurity Professional Program — Indian Council for Technical Research and Development
- Cybersecurity Professional — Hacker School (A Unit of Cartel Software)